

Insurance Risk Prediction

Claim Likelihood & Loss Modeling

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Executive Summary

In the insurance industry, setting the right premium is essential for profitability and competitiveness. If premiums do not reflect actual risk, insurers face adverse selection: overcharging low-risk customers drives them to competitors, while undercharging high-risk customers leads to losses. This project builds predictive models on 39,928 historical auto insurance policy records to estimate two key pricing inputs: loss cost per exposure unit (LC) and historically adjusted loss cost (HALC), as well as claim likelihood (CS). Together, these predictions enable insurers to price policies proportional to each policyholder's true risk, supporting a balanced and profitable portfolio.

The data presents two core modeling challenges: 88.8% of policies have zero claims (severe class imbalance), and claim amounts are highly right-skewed with extreme outliers. Standard models struggle to simultaneously learn which policyholders will claim and how large those claims will be. To address this, we tested three progressively complex approaches for loss prediction: **Tweedie GLMs** as interpretable baselines, tree-based models (**XGBoost**, **LightGBM**) to capture nonlinear interactions, and a **two-stage frequency–severity decomposition** that separates claim probability from claim severity. For claim classification, we evaluated six models ranging from **Logistic Regression** to **CatBoost** and **neural networks**.

The Two-Stage LightGBM model achieved the strongest loss prediction performance (LC MSE: 251,107; HALC MSE: 992,062). By letting a classifier specialize on the zero-inflated structure and a Tweedie regressor focus on severity among claimants, this approach mirrors established actuarial practice and outperformed all alternatives including a three-model stacking ensemble. The same Stage 1 classifier also achieved the highest ROC-AUC for claim classification (0.8586), outperforming CatBoost (0.8583) and all other standalone models, enabling a unified pipeline for both tasks.

SHAP analysis and GLM inference jointly identify time since last renewal, net premium (X.14), and driver age as the strongest predictors of both claim likelihood and loss severity. This consistency across tasks validates the unified modeling framework. Vehicle type and policy cancellation history emerge as additional actionable risk signals not typically weighted in standard pricing.

We recommend three actions for implementation. First, insurers should adopt the frequency–severity decomposition for premium setting, as separately modeling who will claim and how much it will cost enables more granular, accurate pricing than aggregate loss prediction. Second, classification thresholds should be tuned below the default 0.5; lowering to 0.3 increases claim recall from 18% to 43%, which is critical given that missing high-risk policyholders carries greater financial cost than overestimating risk for safe ones. Third, renewal timing and payment behavior (policy cancellations) should be incorporated into risk scoring alongside traditional demographic and vehicle factors, as these variables show strong predictive power but are underutilized in conventional actuarial models. Extreme tail losses remain inherently difficult to predict at the individual level and should continue to be managed through portfolio-level reserves.

1. Introduction

Insurance pricing is a critical challenge in the insurance industry, as insurers must accurately estimate expected losses to set appropriate premiums. Incorrect pricing can lead to adverse selection, where low-risk customers leave and high-risk customers remain, resulting in financial losses. Therefore, accurately predicting both claim likelihood and claim severity is essential for maintaining a balanced and profitable portfolio.

In this project, we use historical policy data to predict key insurance risk metrics. We focus on two main tasks: predicting loss cost per exposure unit (LC) and historically adjusted loss cost (HALC), as well as predicting claim status (CS), which indicates whether a policyholder will make a claim. Together, these tasks provide a framework for modeling both the frequency and severity of claims.

However, the data presents significant challenges. The classification task suffers from severe class imbalance, while the loss variables are highly skewed with many zero values and a few extreme observations. To address these issues, we apply and compare multiple statistical and machine learning models designed to handle imbalanced and skewed data, aiming to improve prediction accuracy and generate insights for insurance risk management.

2. Data Description and Preprocessing

2.1 Data Overview

The training set contains 39,928 policy records across 28 variables; the test set contains 13,310 records with claim-related columns (X.15, X.16, X.17, X.18) withheld. Variables fall into three groups: policy/customer features (contract dates, tenure, number of policies, payment method, net premium), vehicle features (type, horsepower, market value, fuel type, weight), and risk indicators (urban/rural, single/multiple drivers). Six variables are categorical (X.7, X.13, X.19, X.20, X.21, X.27); the remainder are numeric. Key predictors include X.14 (net premium, reflecting existing risk pricing), X.19 (vehicle type, with motorbikes showing higher claim rates), and X.20 (urban/rural location).

2.2 Preprocessing Steps

Five raw date columns were converted into six numeric features - driver age, license years, age at license, policy tenure, time since last renewal, and vehicle age, using November 30, 2018 as the reference date (the latest confirmed event date in the dataset). Three data quality issues were corrected: 35 records with renewal dates preceding policy start dates, 453 zero-horsepower records imputed with vehicle-type group medians, and seven implausible driver ages capped at 85. X.27 (fuel type) missing values were filled with the training mode and binary encoded; five categorical variables were converted to category dtype for native tree model handling. To prevent data leakage, X.15 and X.16 (target components) and X.17 and X.18 (absent from test set) were excluded from the feature matrix, leaving 23 predictors. The training data was split 80/20 (31,942 train / 7,986 validation) with all imputation constants derived from the training set only. These preprocessing steps ensure a consistent feature set for both Task 1 and Task 2 and prevent data leakage during model training.

3. Modeling Approach

Task 1: Loss Prediction (LC and HALC)

We evaluated three families of models for predicting LC and HALC:

- **GLM / Tweedie-based models:** Tweedie GLM and Ridge-regularized GLM served as interpretable baselines with statistical significance, but cannot capture nonlinear interactions.

- **Tree-based models (XGBoost, LightGBM):** Trained with Tweedie objectives (variance power = 1.5), these reduced LC MSE to ~253,400 by capturing nonlinear feature interactions.
- **Two-stage modeling approach:** Since 88.8% of policies have zero claims, we separated the problem into a classifier (claim probability) and a Tweedie regressor (severity among claimants only). This consistently outperformed all direct models and a stacking ensemble.

Final Model Selection

We selected the Two-Stage LightGBM model for both LC and HALC. A LightGBM classifier first predicts claim probability, then a LightGBM Tweedie regressor predicts severity among claimants only, combined as $E[\text{Loss}] = P(\text{claim}) \times E[\text{severity} | \text{claim}]$. This decomposition lets each sub-model specialize on the zero-inflated, right-skewed claim distribution. LightGBM edged out XGBoost in both MSE and training speed. SHAP analysis confirmed the top drivers: *time_since_last_renewal*, *net premium (X.14)*, and *driver_age*, are actuarially interpretable.

Model Performance

Model	LC MSE	HALC MSE
GLM Tweedie (statsmodels)	257,462	1,016,123
Ridge Tweedie GLM	255,530	1,014,256
XGBoost Tweedie (direct)	253,432	1,002,833
LightGBM Tweedie (direct)	253,491	999,292
Stacking (GLM+XGB+LGB)	251,614	994,860
Two-Stage XGBoost	251,211	992,119
Two-Stage LightGBM	251,107	992,062

Task 2: Claim Likelihood Prediction (CS)

We evaluated six classification models for predicting claim likelihood (CS):

- **Logistic Regression (baseline model):** Logistic Regression serves as an interpretable baseline model, assuming a linear relationship between the input features and the log-odds of making a claim. While simple and easy to interpret, it is limited in capturing nonlinear interactions among variables.
- **Random Forest:** Random Forest is an ensemble learning method that captures nonlinear relationships and feature interactions. By aggregating multiple decision trees, it provides more flexible modeling capability compared to linear models, making it well-suited for complex insurance data.
- **HistGradientBoosting:** A histogram-based gradient boosting classifier that bins continuous features for faster training while maintaining strong predictive accuracy on imbalanced data.
- **CatBoost:** A gradient boosting algorithm with native support for categorical features, using ordered boosting to reduce overfitting. Early stopping was applied with AUC as the evaluation metric.
- **XGBoost:** A regularized gradient boosting classifier with L1/L2 penalties. *scale_pos_weight* (~7.9) was used to upweight the minority class, significantly improving claim recall.
- **MLP Neural Network:** A three-layer feedforward network (128-64-32 neurons, ReLU activation) with standardized inputs and early stopping. Captures smooth, nonlinear boundaries.

Final Model Selection

CatBoost achieved the highest standalone ROC-AUC (0.8583), followed closely by HistGradientBoosting (0.8565) and XGBoost (0.8508), with all boosting models outperforming Random Forest (0.8472), MLP (0.8191), and Logistic Regression (0.7518). However, the **LightGBM classifier** embedded in the Task 1 two-stage pipeline achieved **0.8586**, marginally outperforming all standalone models. Since this classifier already produces claim probabilities as part of the loss prediction pipeline, we adopt it as the final CS model. This unifies both tasks under a single two-stage architecture: Stage 1 provides CS predictions for Task 2, and Stage 1 × Stage 2 provides LC and HALC predictions for Task 1. The Task 2 model comparison validates this choice by confirming the LightGBM classifier matches or exceeds every alternative tested.

Model Performance

Model	ROC-AUC
CatBoost	0.8583
HistGradientBoosting	0.8565
XGBoost	0.8508
Random Forest	0.8472
MLP Neural Network	0.8191
Logistic Regression	0.7518

At threshold 0.5, most models predict nearly all cases as no-claim due to class imbalance, yielding low recall (~18%). Lowering the threshold to 0.3 improves claim recall to ~43% at the cost of more false positives. In insurance, missing high-risk policyholders is costlier than false alarms, so threshold tuning is essential for deployment.

4. Model Interpretation

We analyze feature importance from LightGBM alongside SHAP values to identify key drivers of both claim likelihood (CS) and loss severity (LC and HALC). For claim likelihood (CS), feature importance from the LightGBM classifier identifies net premium (X.14), time_since_last_renewal, and driver_age as the strongest predictors of whether a claim occurs. Policy cancellations due to nonpayment (X.12) and vehicle type (X.19) also contribute significantly, reflecting that both customer behavior and inherent risk characteristics drive claim probability. These findings are consistent across all gradient boosting classifiers tested, confirming that the key risk signals are robust to model choice.

For loss severity (LC and HALC), SHAP analysis on the gradient boosting models identifies time_since_last_renewal, net premium (X14), and driver age as the most influential features. Higher premiums are associated with higher predicted losses, consistent with actuarial pricing practices. Younger drivers tend to have higher SHAP values, indicating greater expected loss. Additionally, vehicle characteristics such as horsepower (X23) and market value (X25) increase predicted severity, reflecting higher repair or replacement costs.

Same core variables dominate across all three targets, suggesting that the historical adjustment factor (X.18) scales the outcome rather than changing the underlying risk drivers. These results align with industry expectations to segment customers by risk and design targeted pricing strategies. SHAP further enhances transparency by making predictions explainable, supporting regulatory compliance.

5. Business Insights

Insurance risk is driven by complex, nonlinear feature interactions. Tree-based models consistently outperformed the Tweedie GLM (LC MSE: 253,432 vs. 257,462), while the two-stage frequency–severity decomposition reduced error by letting each sub-model specialize on the zero-inflated distribution. We retain the GLM alongside these models for its interpretable coefficients, which support the regulatory transparency insurers require. The 88.8% zero-claim rate and heavy right skew in loss amounts further validate separately pricing claim frequency and severity for more granular premium setting.

A key practical consideration is classification threshold tuning. At the default 0.5 cutoff, gradient boosting classifiers achieve high accuracy but low claim recall (~18%), since the model defaults to predicting non-claim for most policies. Lowering the threshold improves claim detection at the cost of more false positives, a tradeoff insurers should calibrate based on risk appetite, given that underpricing a high-risk policyholder is costlier than slightly overcharging a safe one.

Across both tasks, the same features dominate: time since last renewal, net premium (X.14), driver age, and cancellation history (X.12). Renewal timing and payment behavior represent underutilized signals that could strengthen risk segmentation beyond traditional demographic and vehicle factors. These models are trained on auto insurance data and would require re-training for other lines.

6. Conclusion

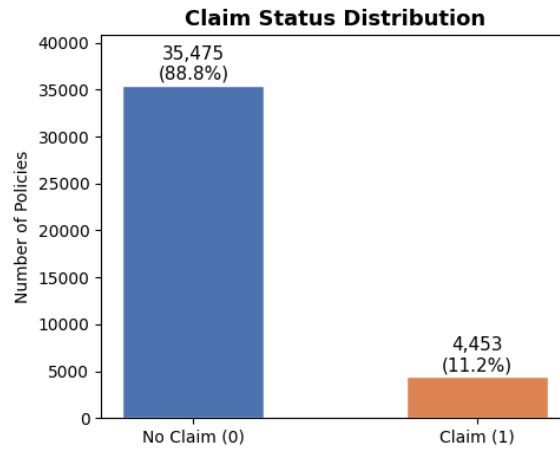
This project developed a unified predictive modeling framework to address two core insurance pricing tasks: estimating loss cost per exposure unit (LC) and historically adjusted loss cost (HALC), and predicting claim likelihood (CS). For loss prediction, the Two-Stage LightGBM model was the best-performing approach. For claim classification, CatBoost achieved the highest ROC-AUC among standalone models (0.8583); however, the LightGBM classifier embedded in the two-stage pipeline marginally outperformed it at 0.8586, enabling a single unified architecture to serve both tasks simultaneously.

A central finding is that nonlinear modeling is a necessity given the complexity of real-world claim data. Tree-based gradient boosting models consistently outperformed Tweedie GLMs by capturing feature interactions that linear models cannot represent, while remaining interpretable through SHAP analysis. It is also important to separate loss frequency and severity. Since 88.8% of policies carry zero claims, a single aggregate model is forced to simultaneously learn claim incidence and loss magnitude. Decomposing the problem into a classifier that estimates claim probability and a Tweedie regressor that estimates severity among claimants directly mirrors established actuarial practice. Finally, classification threshold tuning was essential for practical deployment. Lowering the decision threshold from 0.5 to 0.3 increased claim recall from 18% to 43% which reflects an asymmetric cost structure.

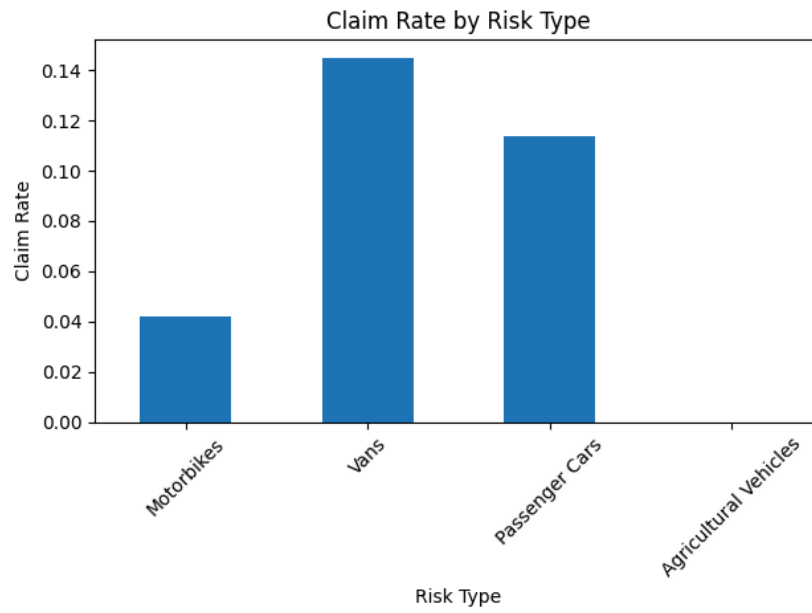
Together, these modeling choices inform how auto premiums should be adjusted. By separately quantifying who is likely to claim and how much that claim will cost, insurers can price policies in proportion to each policyholder's risk profile, reducing adverse selection and supporting long-term portfolio profitability.

Appendix

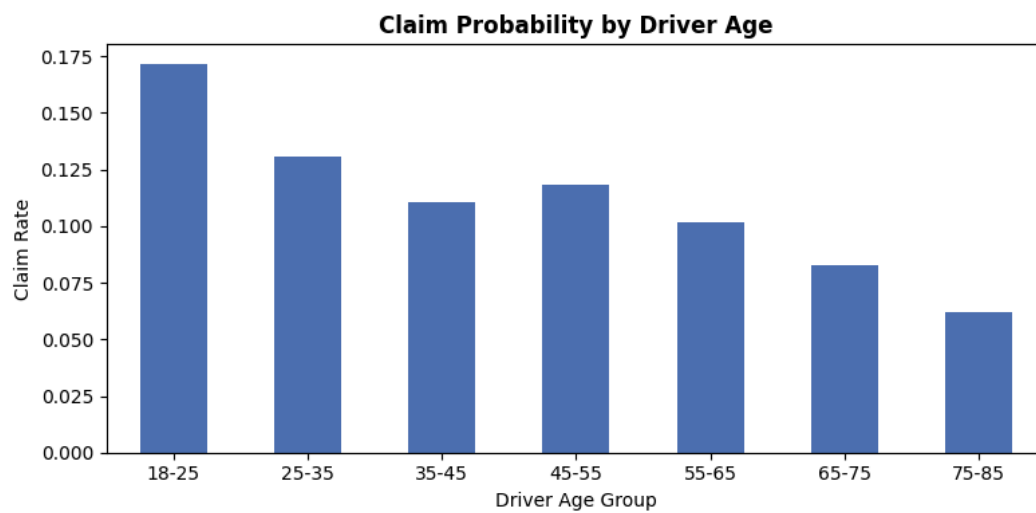
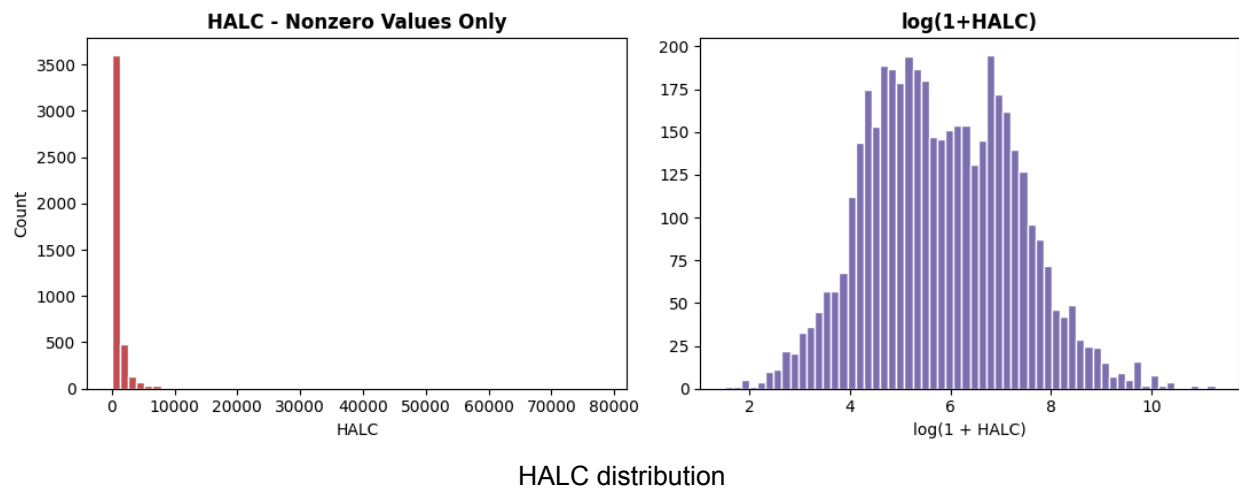
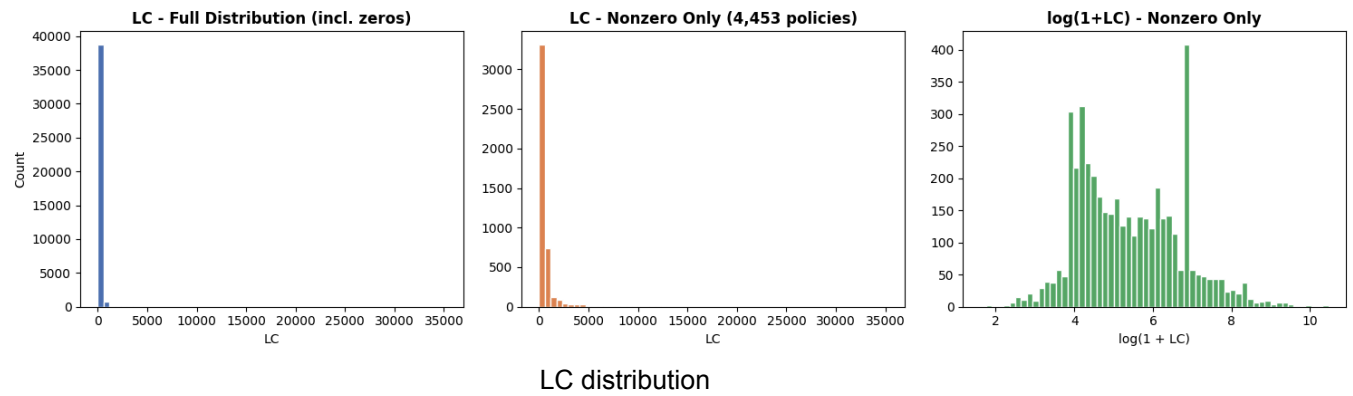
- Additional plots



CS class distribution bar chart



Claim rate by vehicle type



Claim probability by driver age bins

- Hyperparameter tuning results

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--- GLM (LC) Top Coefficients by Magnitude ---
      feature      coef      pvalue
driver_age  36.720489  9.714826e-01
license_years -36.134371  9.713798e-01
age_at_license -17.843289  9.715687e-01
X.19_4    -3.277266  9.247220e-01
const      3.266666  2.603654e-01
time_since_last_renewal  0.853101  2.840550e-57
X.12     -0.665370  4.363460e-28
policy_tenure -0.568686  9.315862e-11
X.19_3     0.490756  4.492851e-03
X.19_2     0.425376  7.397630e-03
X.14       0.278291  9.571351e-11
X.10       0.232071  4.631924e-03
X.13_1     0.227413  5.194637e-07
X.26     -0.102312  2.085840e-01
X.8       -0.095044  2.041257e-01

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Significant features (p<0.05): 8 / 26

Ridge Tweedie GLM alpha tuning ($\alpha = 0.001$ to 10.0, best = 1.0)

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MODEL 2: Ridge-Regularized Tweedie GLM (sklearn)
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alpha= 0.001 → LC MSE: 257,453.42
alpha= 0.01  → LC MSE: 257,397.70
alpha= 0.1   → LC MSE: 256,907.41
alpha= 1.0   → LC MSE: 255,530.44
alpha= 10.0  → LC MSE: 259,319.02

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Best alpha for LC: 1.0 → MSE: 255,530.44
Best alpha for HALC: 1.0 → MSE: 1,014,255.73

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GLM coefficient table with p-values (top 15 features)

- Additional model comparisons

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TASK 1 MODEL COMPARISON – VALIDATION MSE
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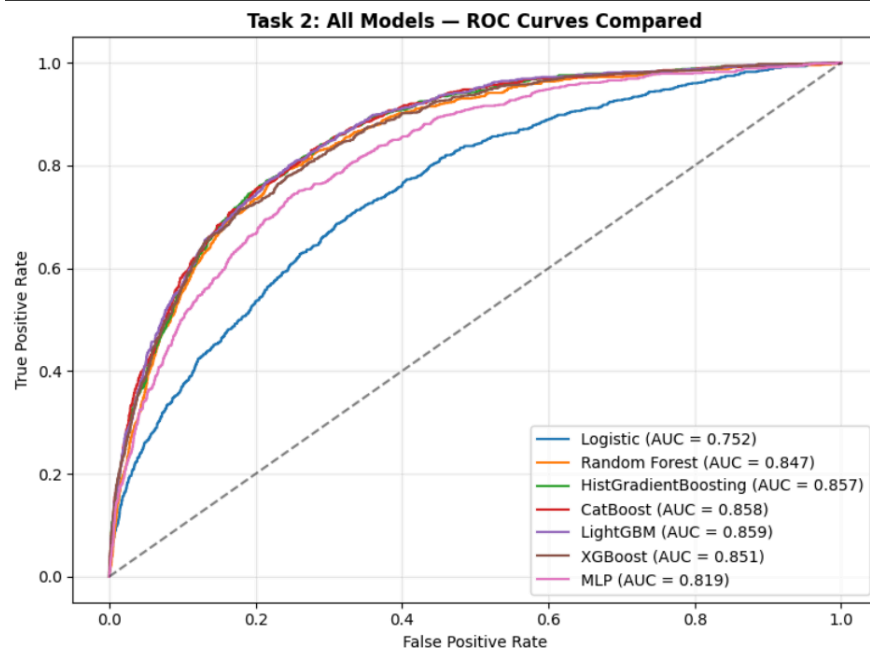
Model	LC_MSE	HALC_MSE	LC_RMSE	HALC_RMSE
XGBoost Tweedie (direct)	253,431.83	1,002,832.73	503.42	1,001.42
Two-Stage XGBoost (existing)	251,211.01	992,118.80	501.21	996.05
GLM Tweedie (statsmodels)	257,461.53	1,016,122.91	507.41	1,008.03
Ridge Tweedie GLM (sklearn)	255,530.44	1,014,255.73	505.50	1,007.10
LightGBM Tweedie (direct)	253,490.57	999,291.75	503.48	999.65
Two-Stage LightGBM	251,106.82	992,062.12	501.11	996.02
Stacking (GLM+XGB+LGB)	251,613.50	994,860.07	501.61	997.43

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Best LC model: Two-Stage LightGBM
Best HALC model: Two-Stage LightGBM

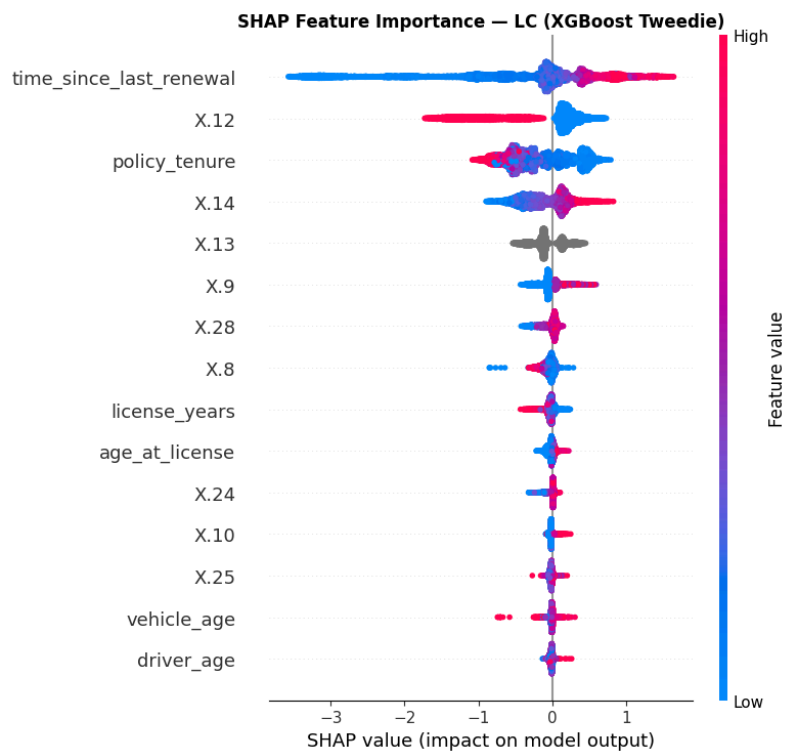
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Task 1 LC/HALC MSE comparison bar chart

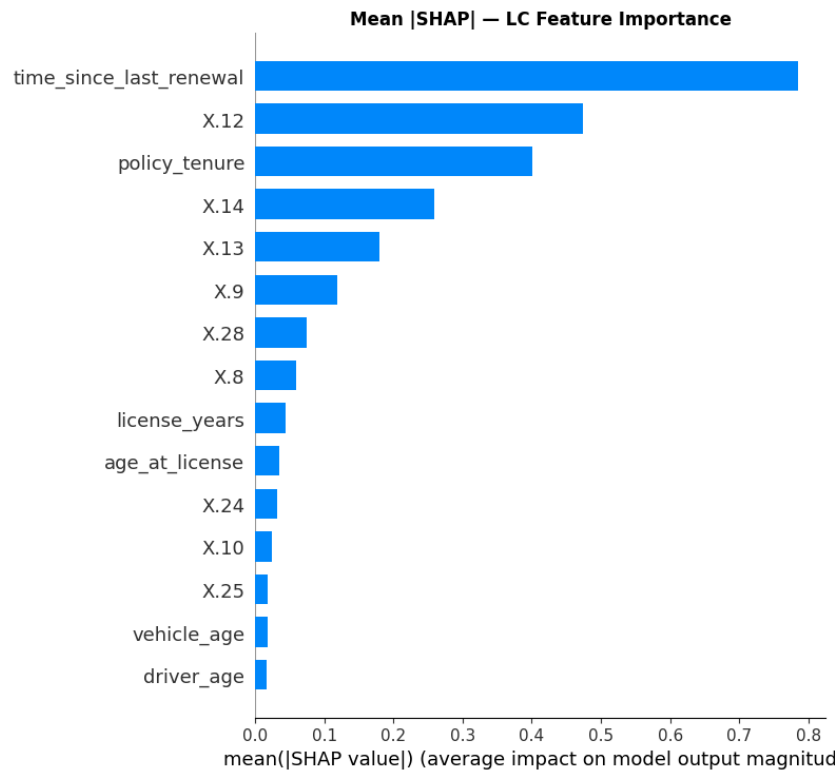


Task 2 combined ROC curves

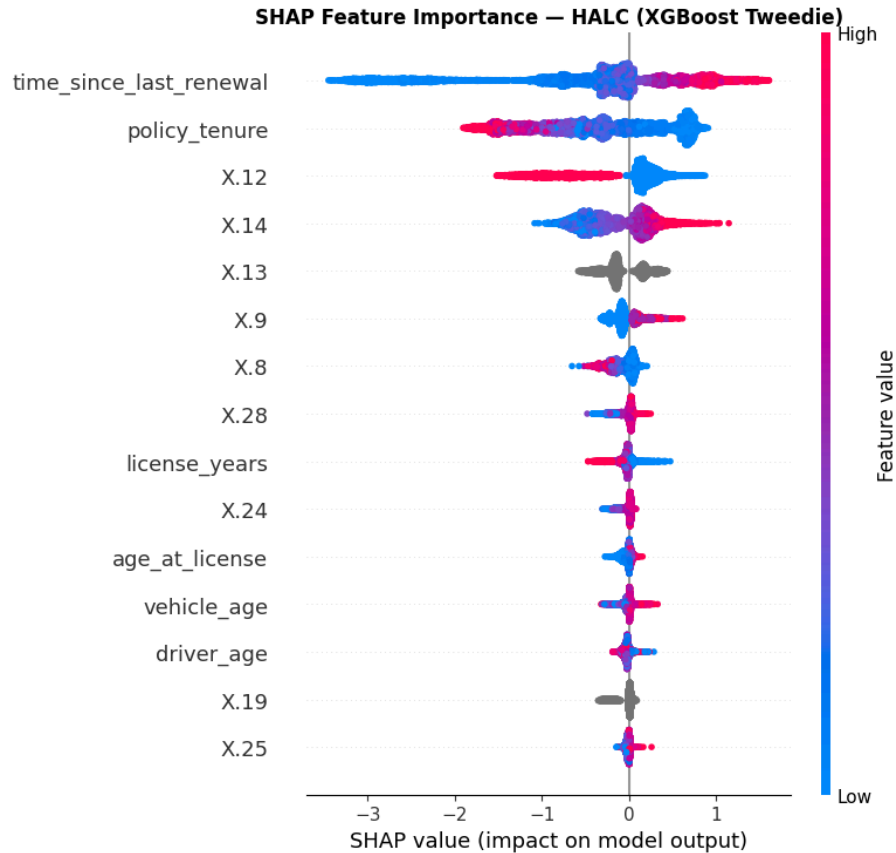
(Noted: *LightGBM classifier (Stage 1): 0.8586 — selected as final model.*)



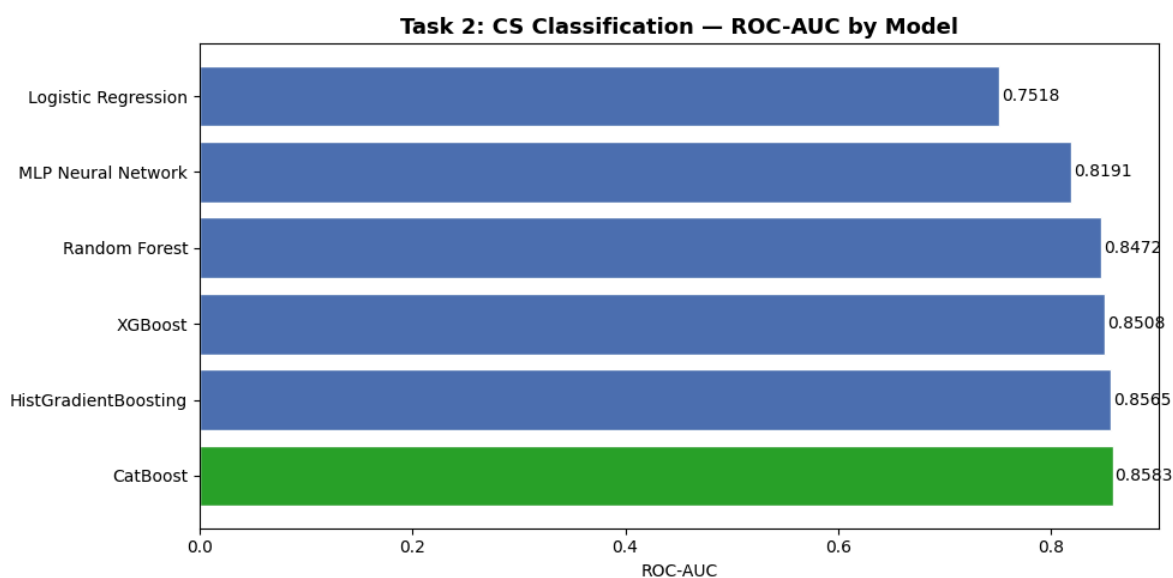
SHAP bee swarm plot — LC



SHAP bar plot — LC mean absolute values



SHAP bee swarm plot — HALC



ROC-AUC Comparison by Model

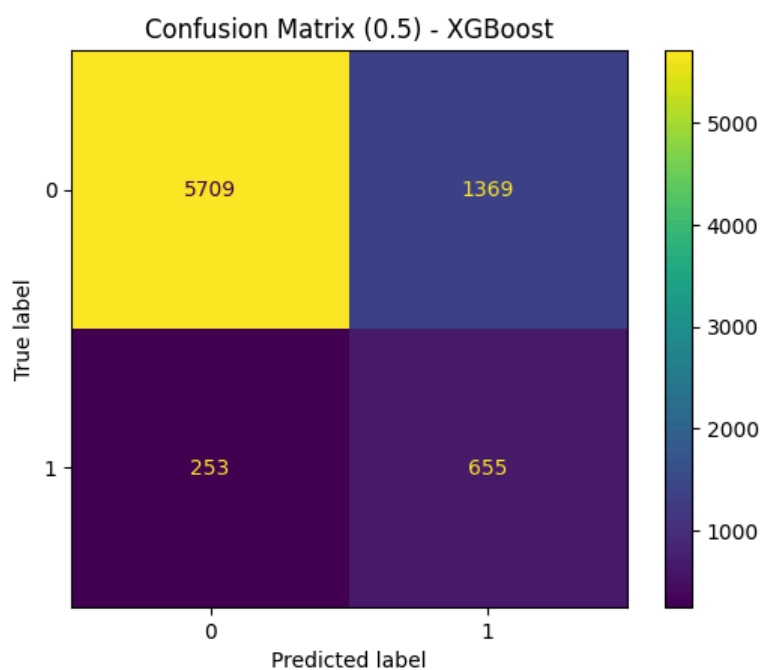


Figure A11: Confusion Matrix — XGBoost (Threshold 0.5)

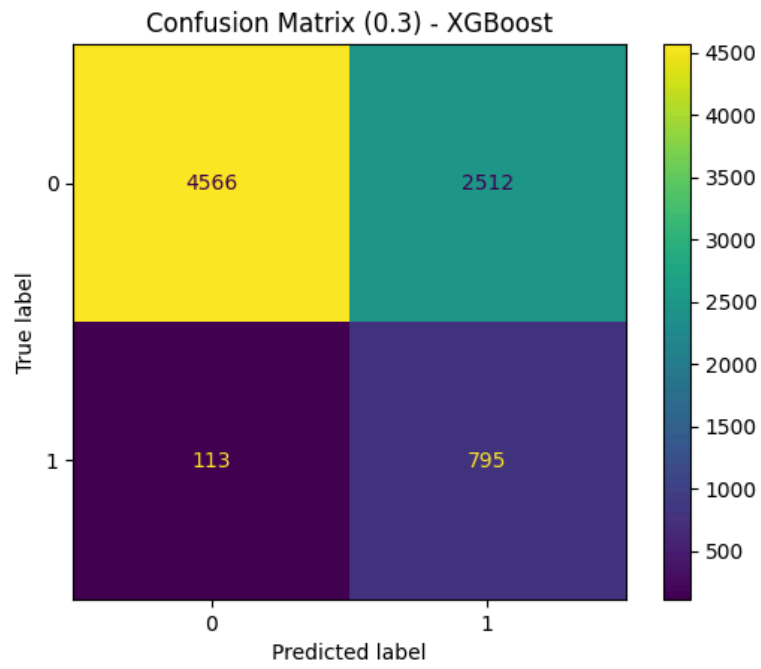


Figure A12: Confusion Matrix — XGBoost (Threshold 0.3)

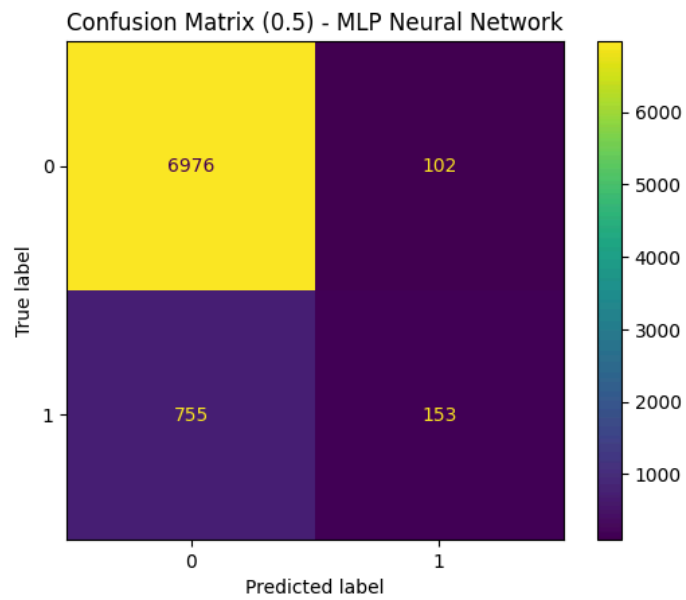


Figure A13: Confusion Matrix — MLP (Threshold 0.5)

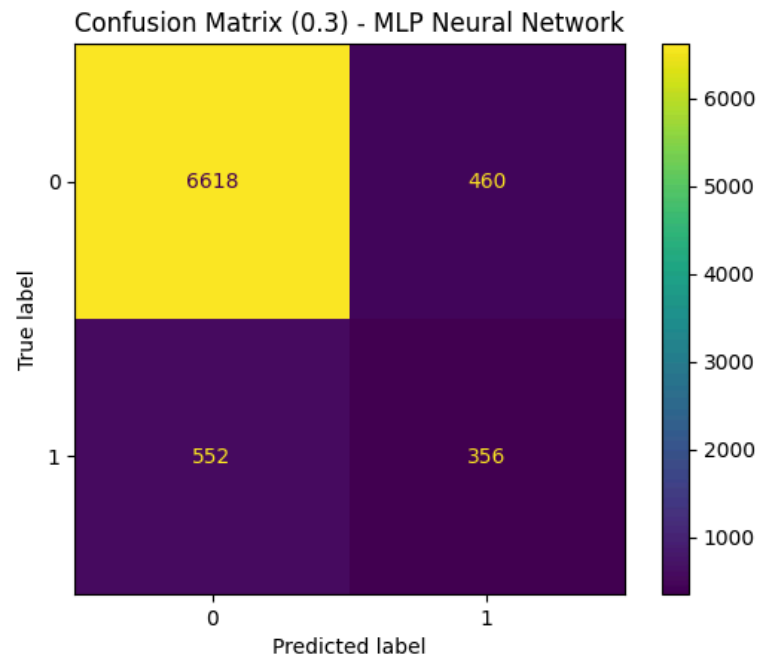


Figure A14: Confusion Matrix — MLP (Threshold 0.3)